

Clustering and Similarity: Retrieving Documents



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Retrieving documents of interest

Document retrieval

- Currently reading article you like



Document retrieval

- Currently reading article you like
- **Goal:** Want to find similar article



Document retrieval



Challenges

- How do we measure similarity?
- How do we search over articles?



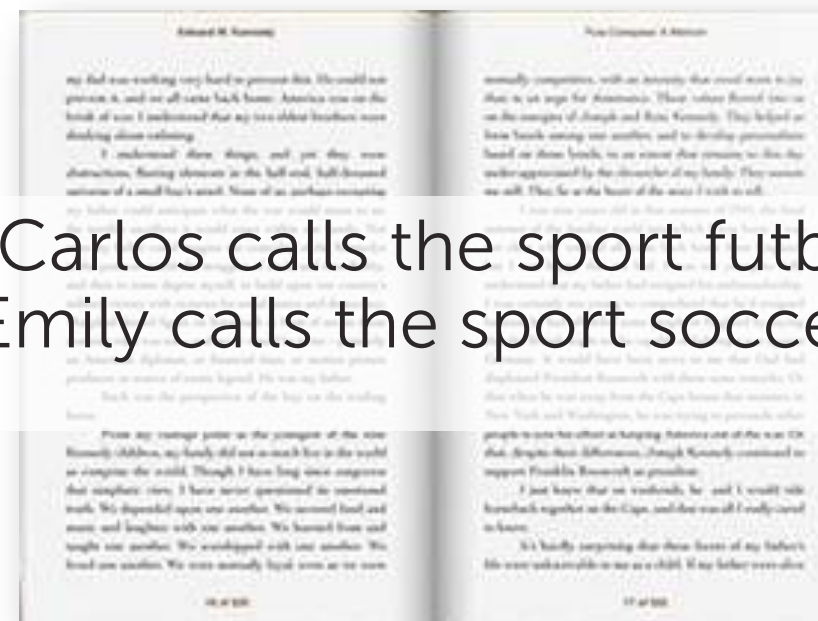
Word count representation for measuring similarity

Word count document representation

- Bag of words model
 - Ignore order of words
 - Count # of instances of each word in vocabulary



“Carlos calls the sport futbol.
Emily calls the sport soccer.”



Measuring similarity



1	0	0	0	5	3	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---



3	0	0	0	2	0	0	1	0	1	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

$$\begin{aligned} &1 \times 3 \\ &+ \\ &5 \times 2 \\ &= 13 \end{aligned}$$



Measuring similarity



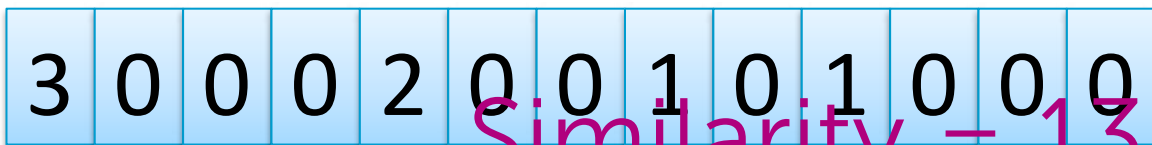
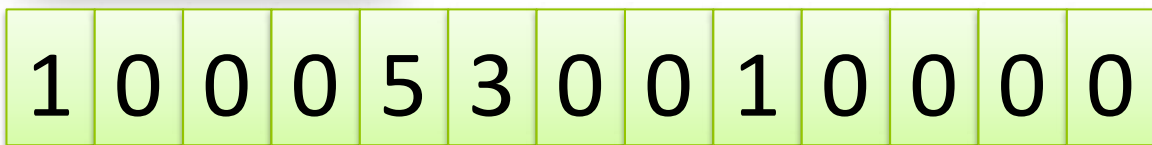
1	0	0	0	5	3	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---



0	0	1	0	0	0	9	0	0	6	0	4	0
---	---	---	---	---	---	---	---	---	---	---	---	---



Issues with word counts – Doc length



Similarity = 13



Similarity = 52



Solution = normalize



1	0	0	0	5	3	0	0	1	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

$$\sqrt{(1^2 + 5^2 + 3^2 + 1^2)}$$

1				5	3			1				
/	0	0	0	/	/	0	0	/	0	0	0	0
6				6	6			6				

Prioritizing important words with tf-idf

Issues with word counts – Rare words



Common words in doc: “the”, “player”, “field”, “goal”

Dominates rare words like: “futbol”, “Messi”

Document frequency

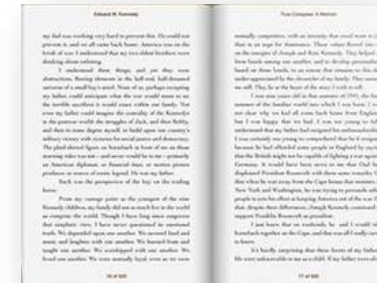
- What characterizes a rare word?
 - Appears infrequently in the corpus
- Emphasize words appearing in few docs
 - Equivalently, discount word w based on
of docs containing w in corpus

Important words

- Do we want only rare words to dominate???
- What characterizes an **important word**?
 - Appears frequently in document (**common locally**)
 - Appears rarely in corpus (**rare globally**)
- Trade off between **local frequency** and **global rarity**

TF-IDF document representation

- Term frequency – inverse document frequency (tf-idf)

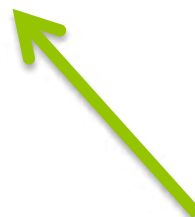


TF-IDF document representation

- Term frequency – inverse document frequency (tf-idf)
- Term frequency



- Same as word counts

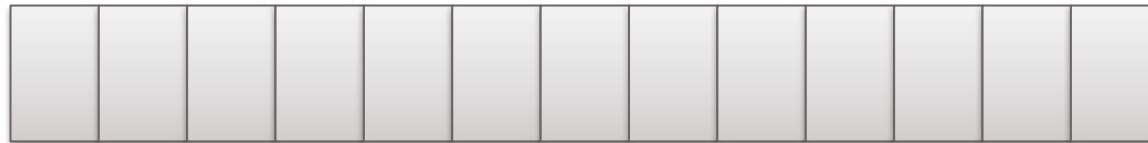


TF-IDF document representation

- Term frequency – inverse document frequency (tf-idf)
- Term frequency



- Inverse document frequency



$$\log \frac{\# \text{ docs}}{1 + \# \text{ docs using word}}$$



TF-IDF document representation

- Term frequency – inverse document frequency (tf-idf)
- Term frequency



- Inverse document frequency



$$\log \frac{\# \text{ docs}}{1 + \# \text{ docs using word}}$$

Handwritten notes explaining the IDF formula:

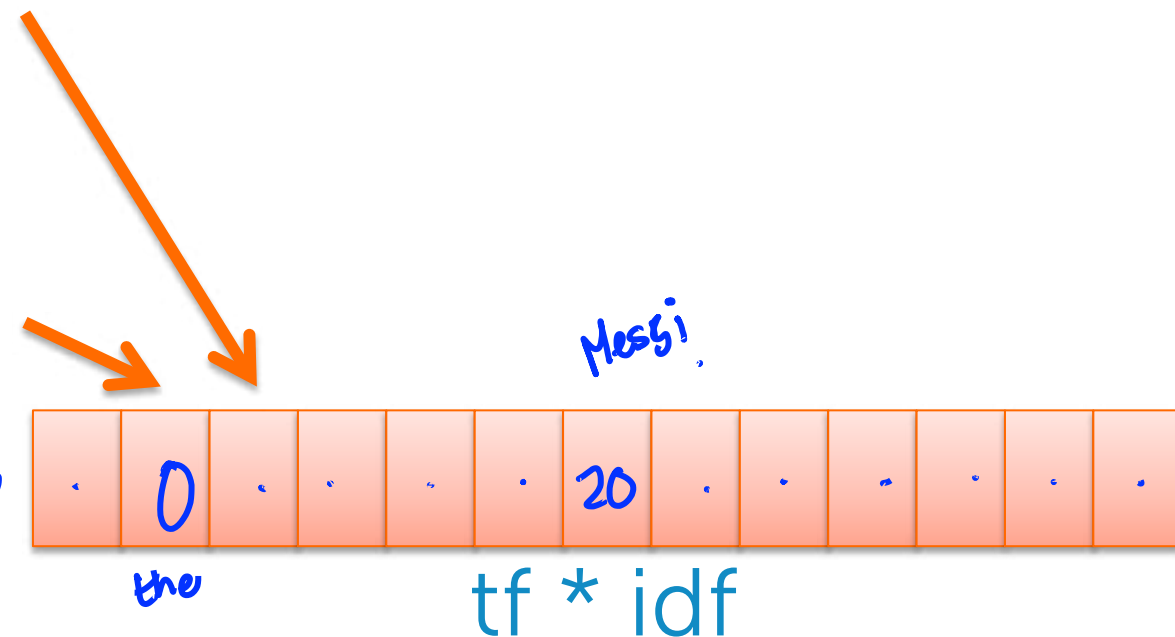
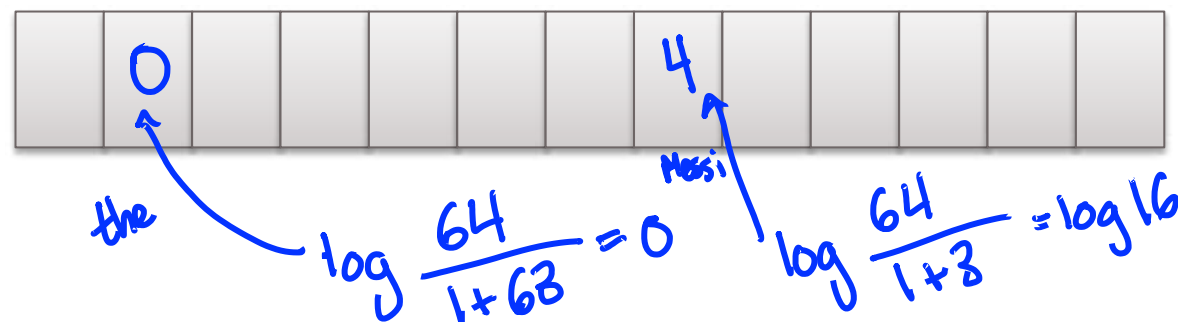
- For a word in many docs: $\log \frac{\text{large \#}}{1 + \text{large \#}} \approx \log 1 = 0$
- For a rare word: $\log \frac{\text{large \#}}{1 + \text{small \#}} \rightarrow \text{large \#}$

TF-IDF document representation

- Term frequency – inverse document frequency (tf-idf)
- Term frequency



- Inverse document frequency



Retrieving similar documents

Nearest neighbor search

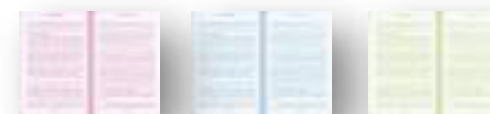
- Query article:



- Corpus:



- **Specify:** Distance metric
- **Output:** Set of most similar articles



1 – Nearest neighbor

- **Input:** Query article 
- **Output:** *Most* similar article
- Algorithm:
 - Search over each article  in corpus
 - Compute $s = \text{similarity}(\text{document icon}, \text{document icon})$
 - If $s > \text{Best}_s$, record  = 
and set $\text{Best}_s = s$
 - Return 

k – Nearest neighbor

- **Input:** Query article 
- **Output:** *List of k* similar articles



Clustering documents

Structure documents by topic

- Discover groups (*clusters*) of related articles



What if some of the labels are known?

- Training set of labeled docs



SPORTS



WORLD NEWS



ENTERTAINMENT



SCIENCE

Multiclass classification problem

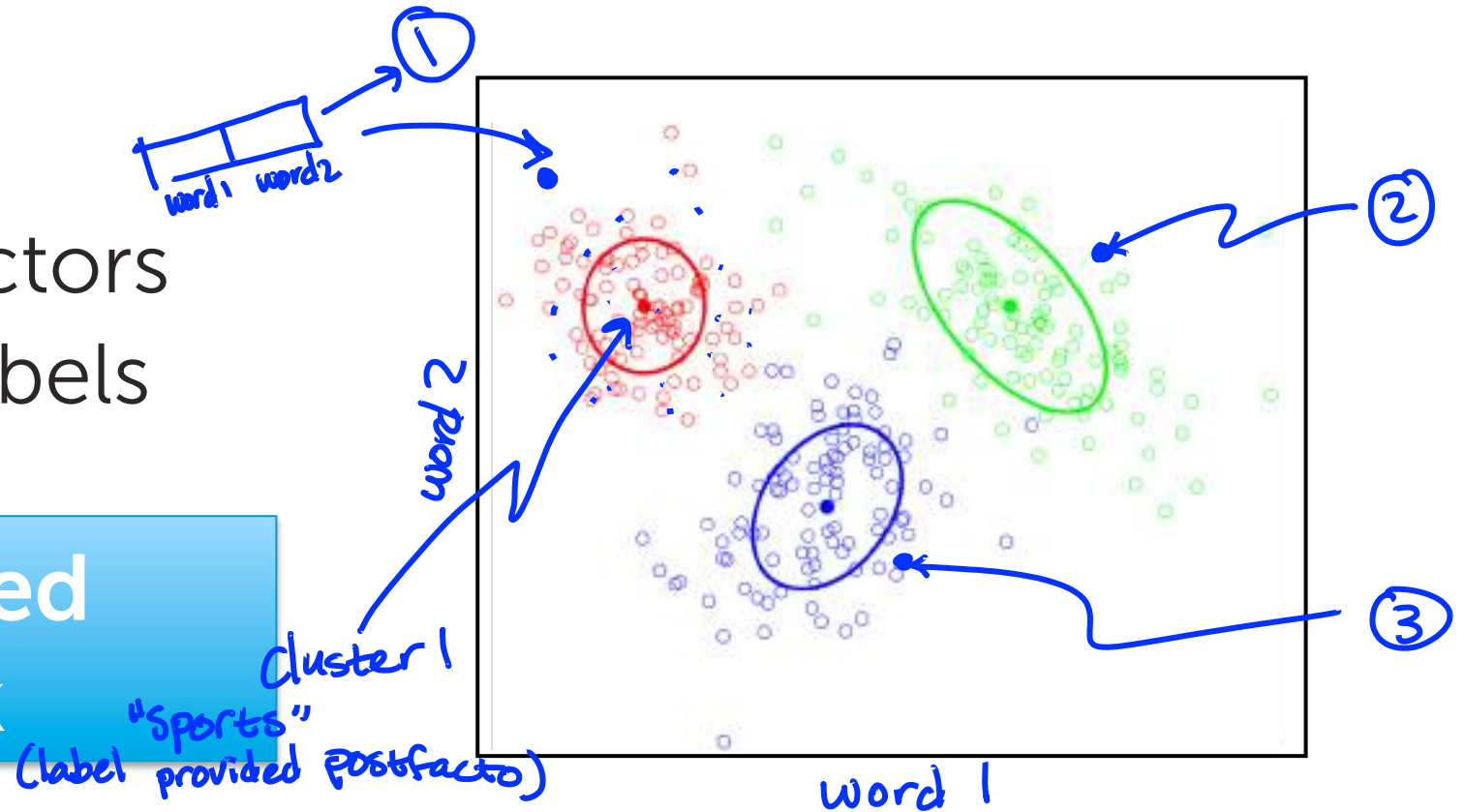


Example of
supervised learning

Clustering

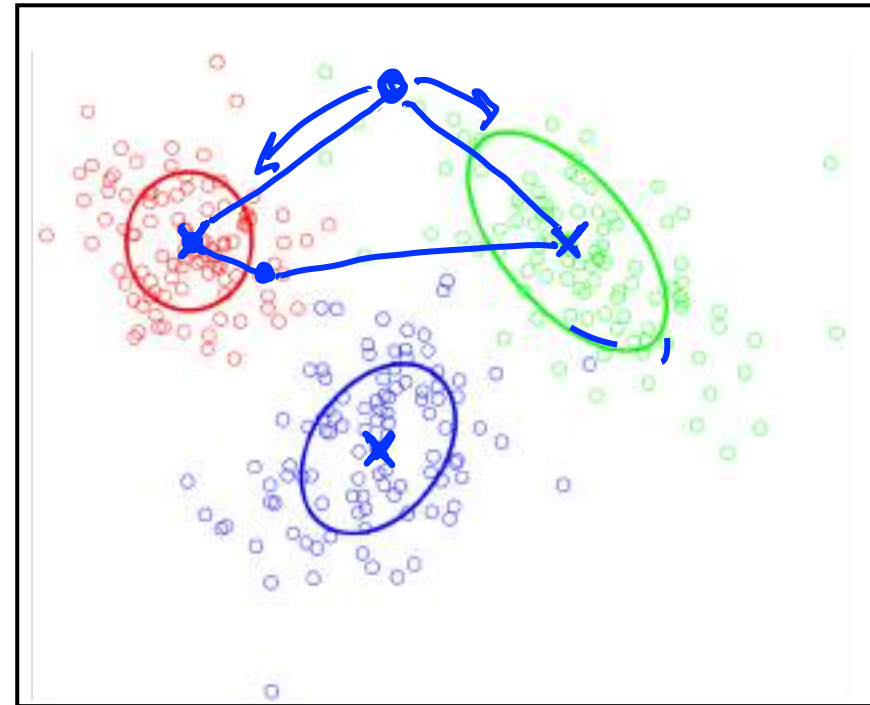
- No labels provided
- Want to uncover cluster structure
- **Input:** docs as vectors
- **Output:** cluster labels

An unsupervised
learning task



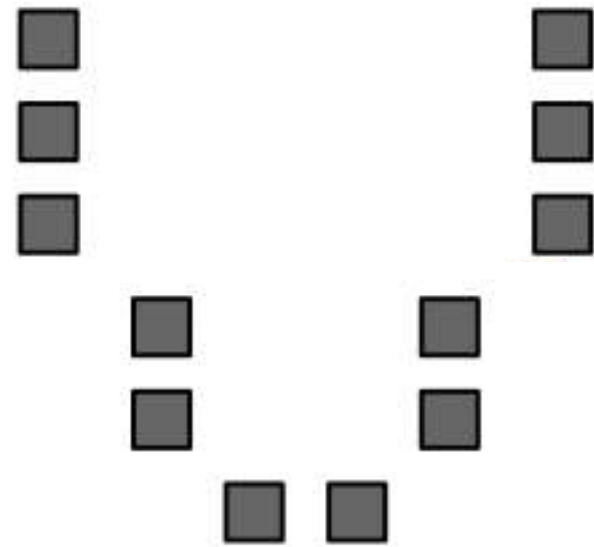
What defines a cluster?

- Cluster defined by **center** & **shape/spread**
- Assign observation (**doc**) to cluster (**topic label**)
 - Score under cluster is higher than others
 - Often, just more similar to assigned cluster center than other cluster centers



k-means

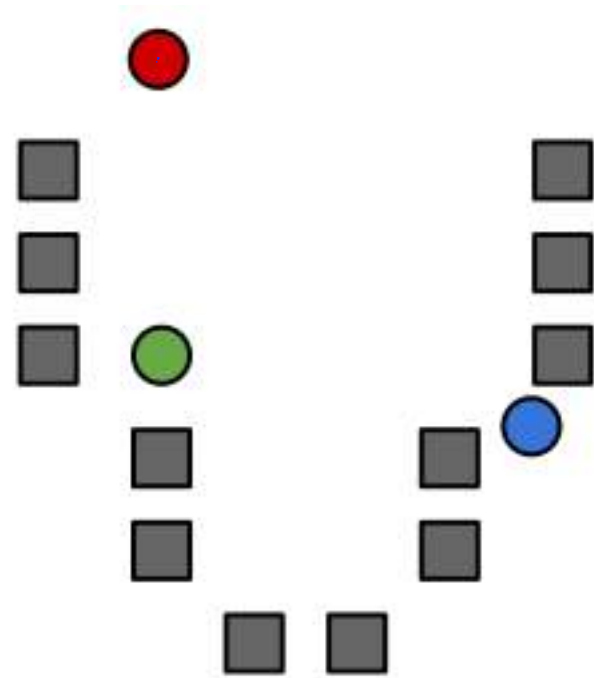
- Assume
 - Similarity metric =
distance to cluster
center
(smaller better)



DATA
to
CLUSTER

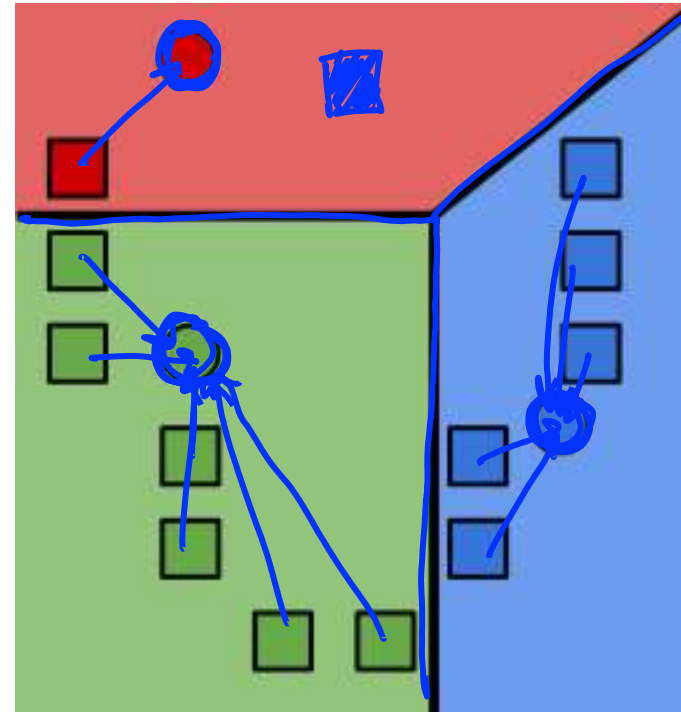
k-means algorithm

0. Initialize cluster centers



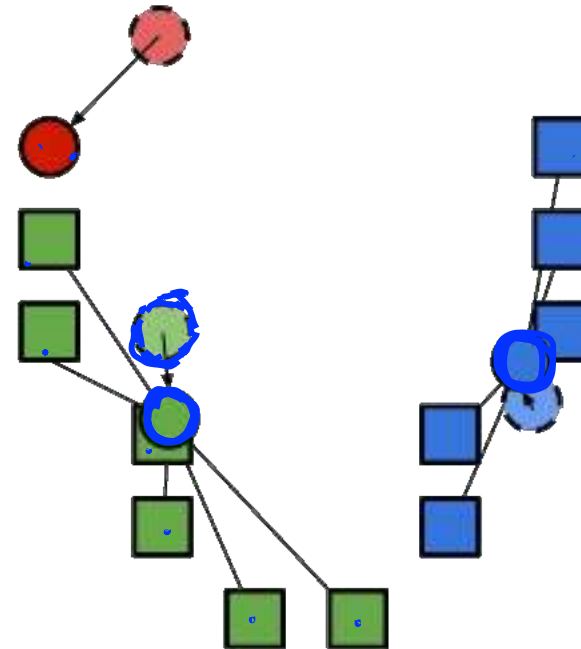
k-means algorithm

0. Initialize cluster centers
1. Assign observations to closest cluster center



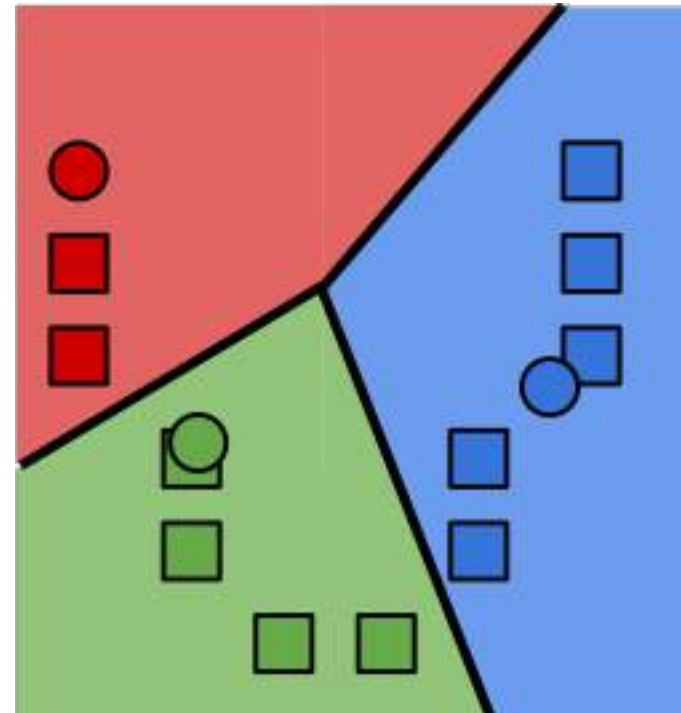
k-means algorithm

0. Initialize cluster centers
1. Assign observations to closest cluster center
2. Revise cluster centers as mean of assigned observations



k-means algorithm

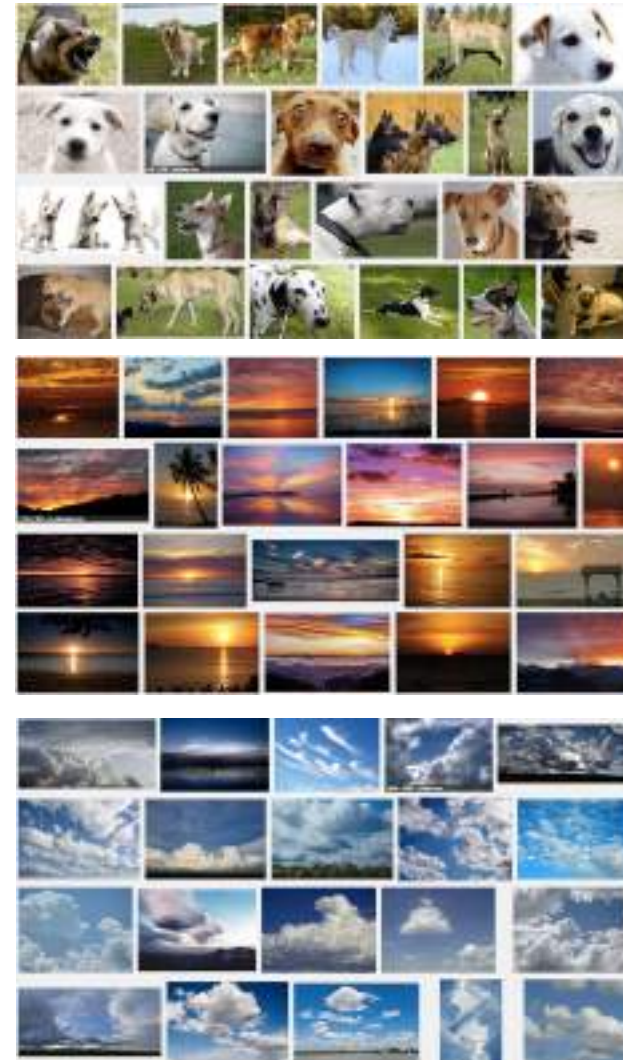
0. Initialize cluster centers
1. Assign observations to closest cluster center
2. Revise cluster centers as mean of assigned observations
3. Repeat 1.+2. until convergence



Other examples

Clustering images

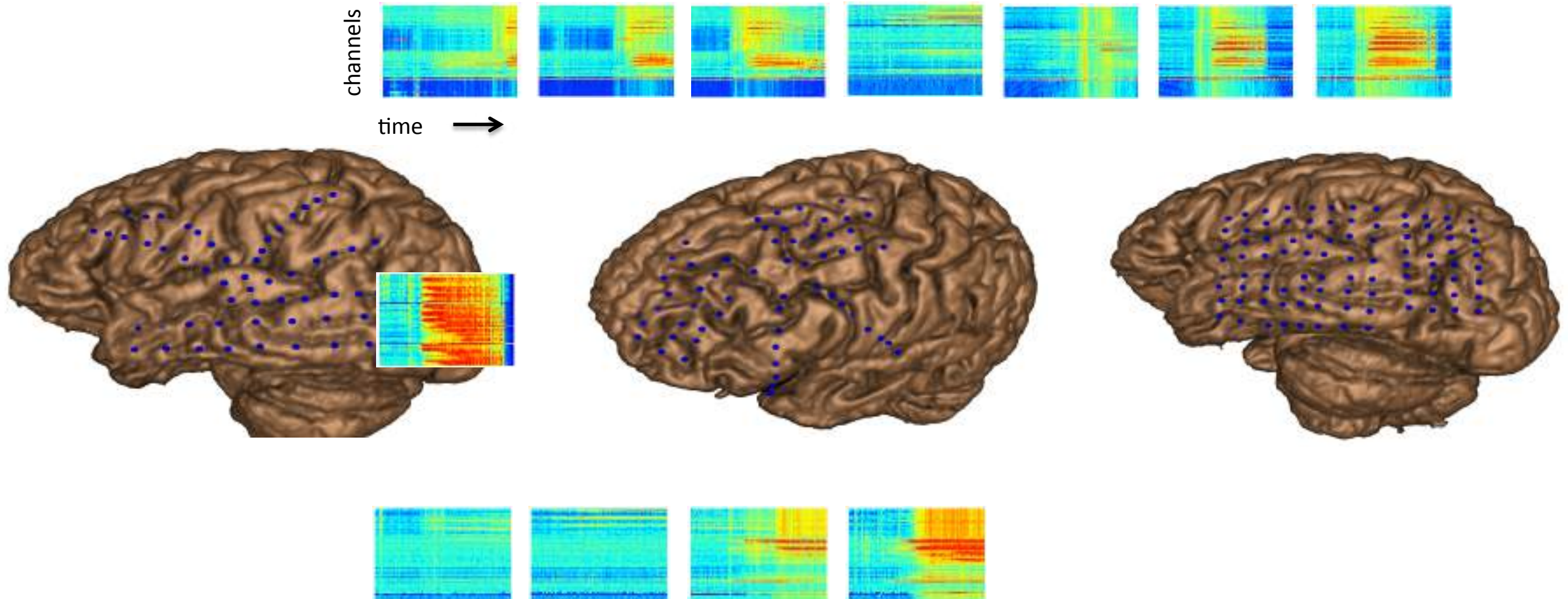
- For search, group as:
 - Ocean
 - Pink flower
 - Dog
 - Sunset
 - Clouds
 - ...



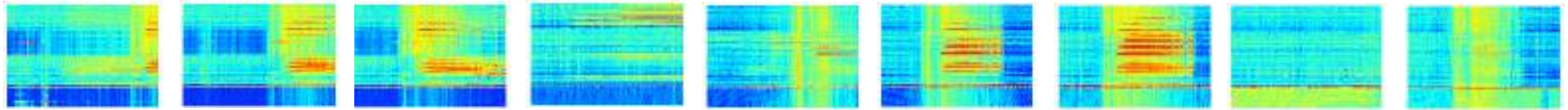
Grouping patients by medical condition

- Better characterize subpopulations and diseases

Example: Patients and seizures are diverse



Cluster seizures by observed time courses



Products on Amazon

- Discover product categories from purchase histories



~~"furniture"~~
"baby"



- Or discovering groups of **users**

Structuring web search results

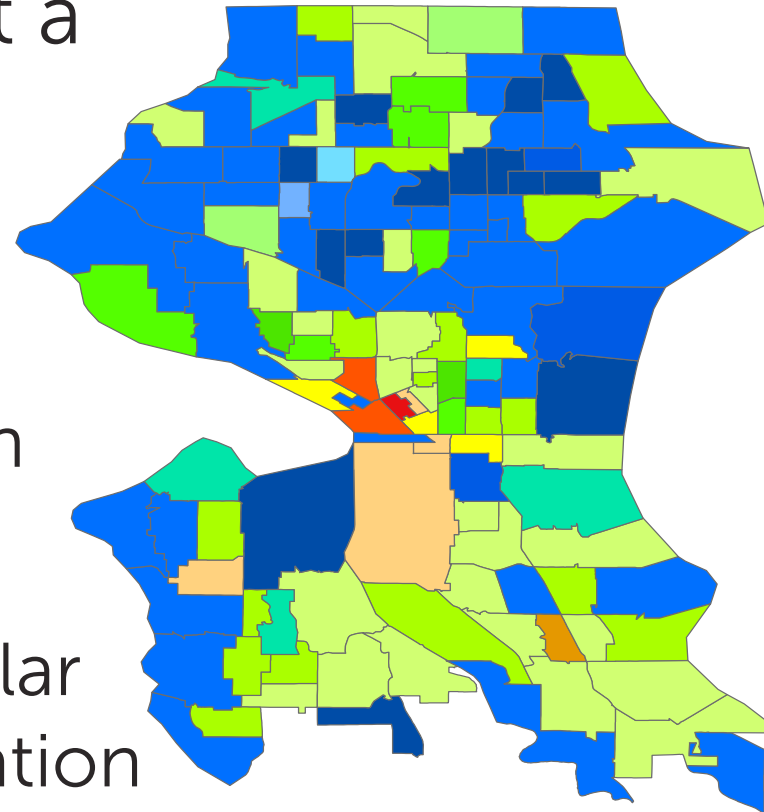
- Search terms can have multiple meanings
- Example: "cardinal"



- Use clustering to structure output

Discovering similar neighborhoods

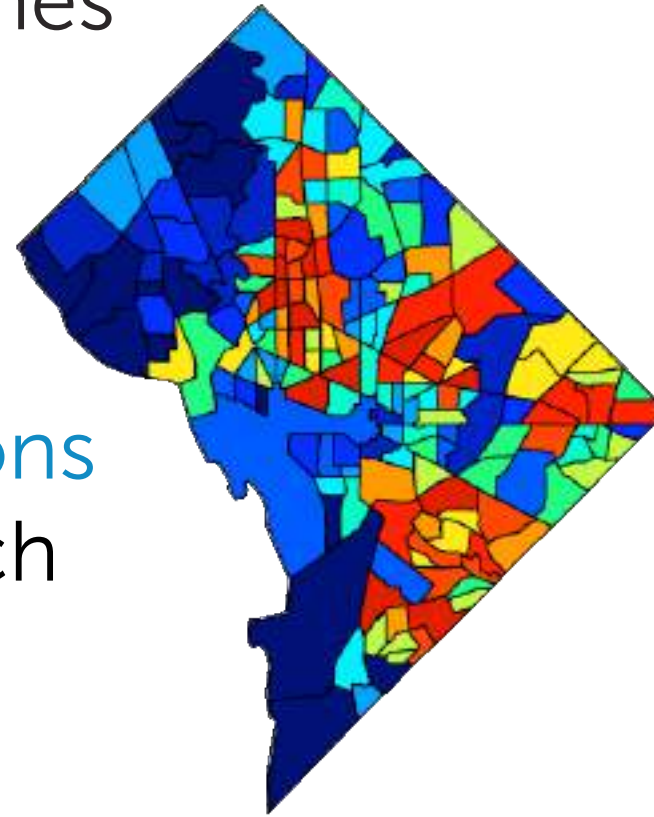
- **Task 1:** Estimate price at a small regional level
- **Challenge:**
 - Only a few (or no!) sales in each region per month
- **Solution:**
 - Cluster regions with similar trends and share information within a cluster



City of Seattle

Discovering similar neighborhoods

- **Task 2:** Forecast violent crimes to better task police
- Again, **cluster regions** and **share information!**
- Leads to **improved predictions** compared to examining each region independently



Washington, DC

Summary for clustering and similarity

What you can do now...

- Describe ways to represent a document (e.g., raw word counts, tf-idf,...)
- Measure the similarity between two documents
- Discuss issues related to using raw word counts
 - Normalize counts to adjust for document length
 - Emphasize important words using tf-idf
- Implement a nearest neighbor search for document retrieval
- Describe the input (unlabeled observations) and output (labels) of a clustering algorithm
- Determine whether a task is supervised or unsupervised
- Cluster documents using k-means (algorithmic details to come...)
- Describe other applications of clustering